

Einführung in die Kern-, Teilchen und Astrophysik

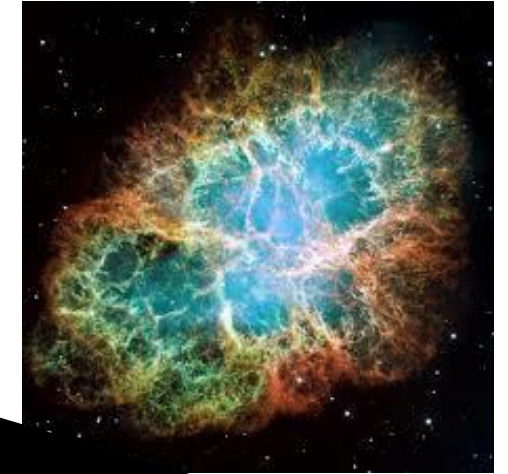
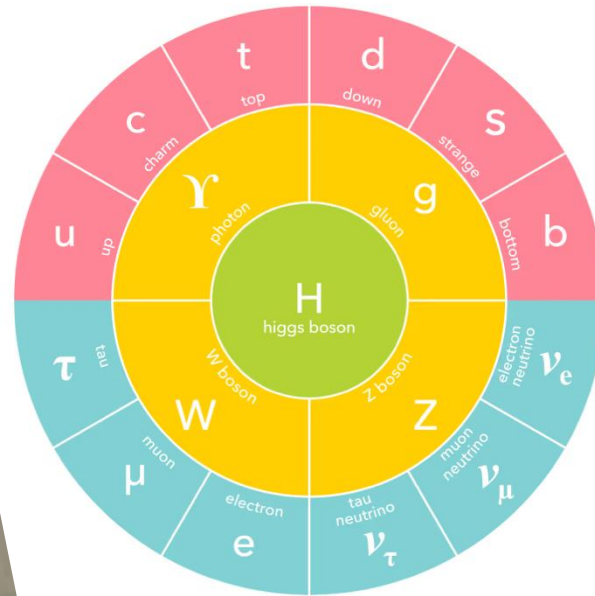
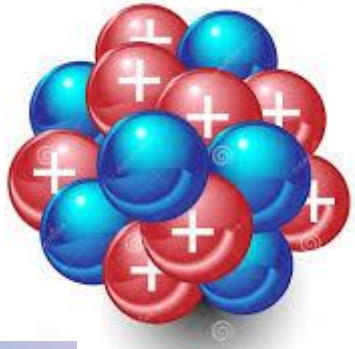


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Lecture	Date	Theme	Topic	Sub-Topics
1	14/10/25	Introduction	Little History of particle physics	atoms, elementary particles
2	21/10/25	Particle Physics	Standard Model	Fermions, Bosons, Interactions,
3	28/10/25		Fundamental symmetries	Conservations, Feynman, Propagator,
4	04/11/25		Fundamental symmetries	Discrete symmetries; Cross section
5	11/11/25		Interaction with Matter	Bethe Bloch, gamma interaction, particle detection
6	18/11/25		Accelerators	Four vector, accelerators, cyclotron
7	25/11/25	Astrophysics	Cosmic accelerators	Supernovae, cosmic rays, BBN
8	02/12/25	Nuclear physics	Basics	Binding energies, Weizäcker
9	09/12/25		Nuclear reactions	Alpha, beta, gamma
10	16/12/25		Nuclear reactors	Radiation, Reactors
11	13/01/26	Particle / Nuclear	High energy physics 1	Deep inelastic scattering
12	20/01/26		High energy physics 2	QCD
13	27/01/26	Particle Physics	Weak interactions	
14	03/02/26	Astrophysics	Neutrinos	



Zusammenfassung der letzten Vorlesung

Noether's Theorem

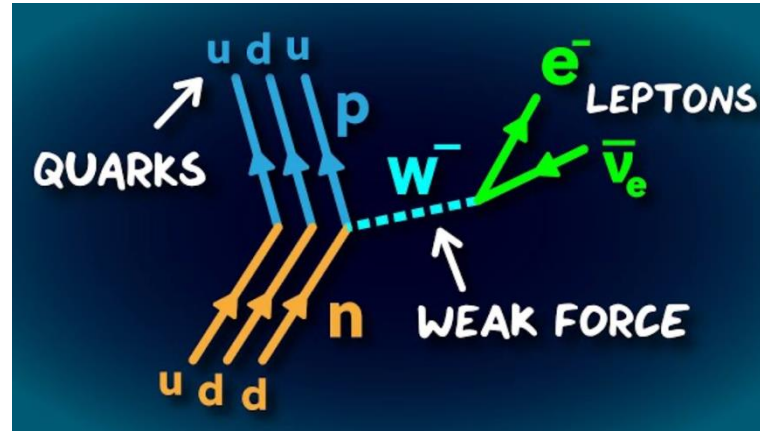
Line Symmetry



Noether's Theorem says if a system has a continuous symmetry, then there must be corresponding quantities whose values are conserved.



Emmy Noether
1882 – 1935



$$\Gamma_{i \rightarrow f} = \frac{2\pi}{\hbar} \left| \langle f | H' | i \rangle \right|^2 \rho(E_f)$$

$$\tilde{V}(\vec{q}) = g_1 g_2 \frac{1}{q^2 + \mu^2}$$

Propagator-Term

Stetige Symmetrien
= Erhaltung

Feynman Graphen

Fermis goldene
Regel

Vereinheitlichung der fundamentalen WW

Jahrhundert

16./17.

18.

Elektrostatik

Magnetismus

Gravitation

19.

Elektromagnetische WW

20.

(1. Hälfte)

QED (Quantenelektrodynamik)

schwache WW

starke WW

ART

20.

(2. Hälfte)

elektroschwache WW

QCD*

Allgemeine
Relativitäts-
theorie

?

Grand Unified Theorie (GUT) ?

„Weltformel“, Theorie of Everything, Stringtheorie?, Quantum Loop Gravity?

Motivation

All nucleons N_B and anti-nucleons $N_{\bar{B}}$ annihilated, and production is no longer possible (universe too cold)

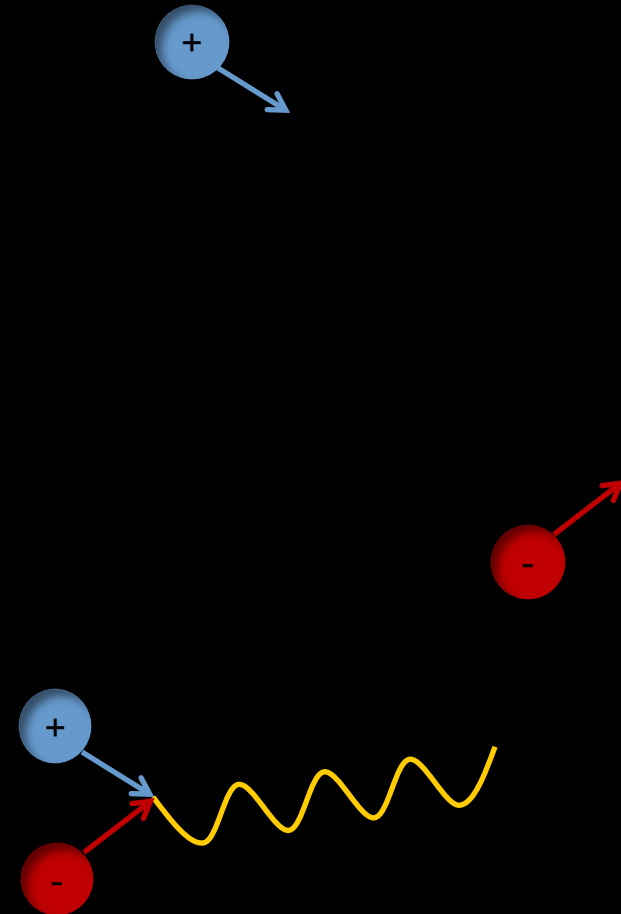
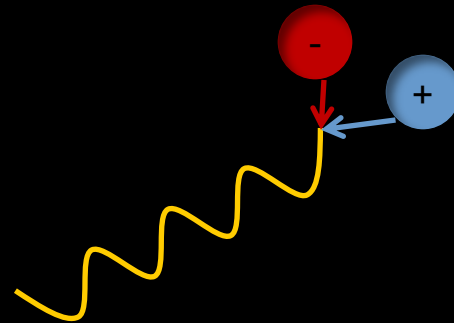
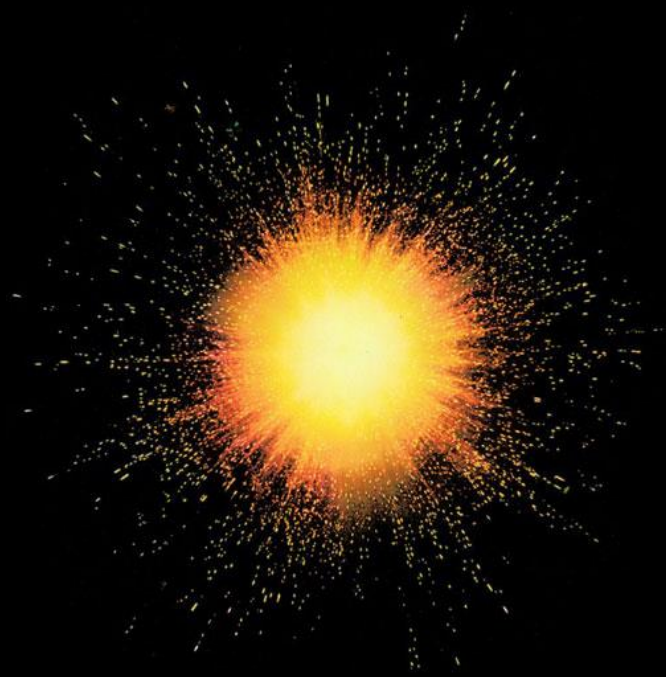
Expected ratio:

$$\frac{N_B}{N_\gamma} = \frac{N_{\bar{B}}}{N_\gamma} = 10^{-18}$$

Observed ratio:

$$\frac{N_B}{N_\gamma} = 10^{-10} \quad \frac{N_{\bar{B}}}{N_\gamma} \approx 0$$

Matter-Antimatter Asymmetry



Sakharov Criteria

$$X \rightarrow qq, \quad X \rightarrow \bar{q}\bar{q}$$

1) **Baryon-number-violating** interactions:

- Process that creates only particles (no antiparticles) and only anti-particles (no particles)
- Lepton number violation can be translated to Baryon number violation (sphaleron processes)

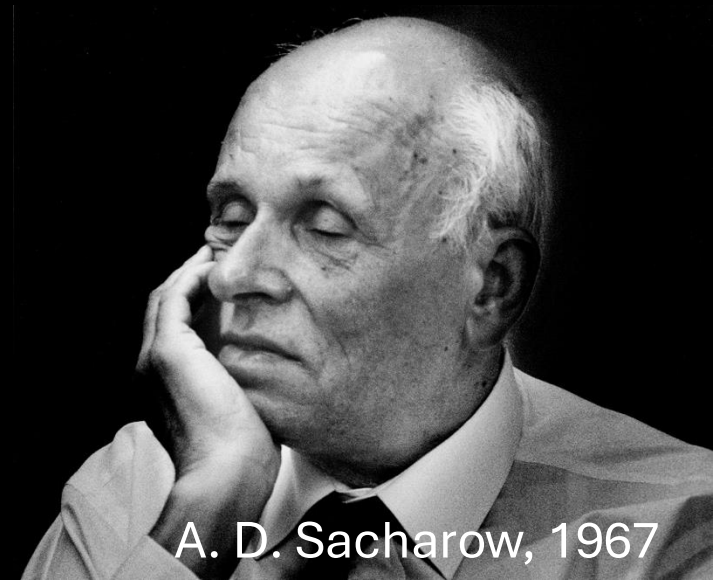
2) **C and CP violation:**

- In addition to having a reaction that creates baryons or anti-baryons only...
- ...we need a difference in the production of baryons over anti-baryons

$$\Gamma(X \rightarrow qq) \neq \Gamma(\bar{X} \rightarrow \bar{q}\bar{q})$$

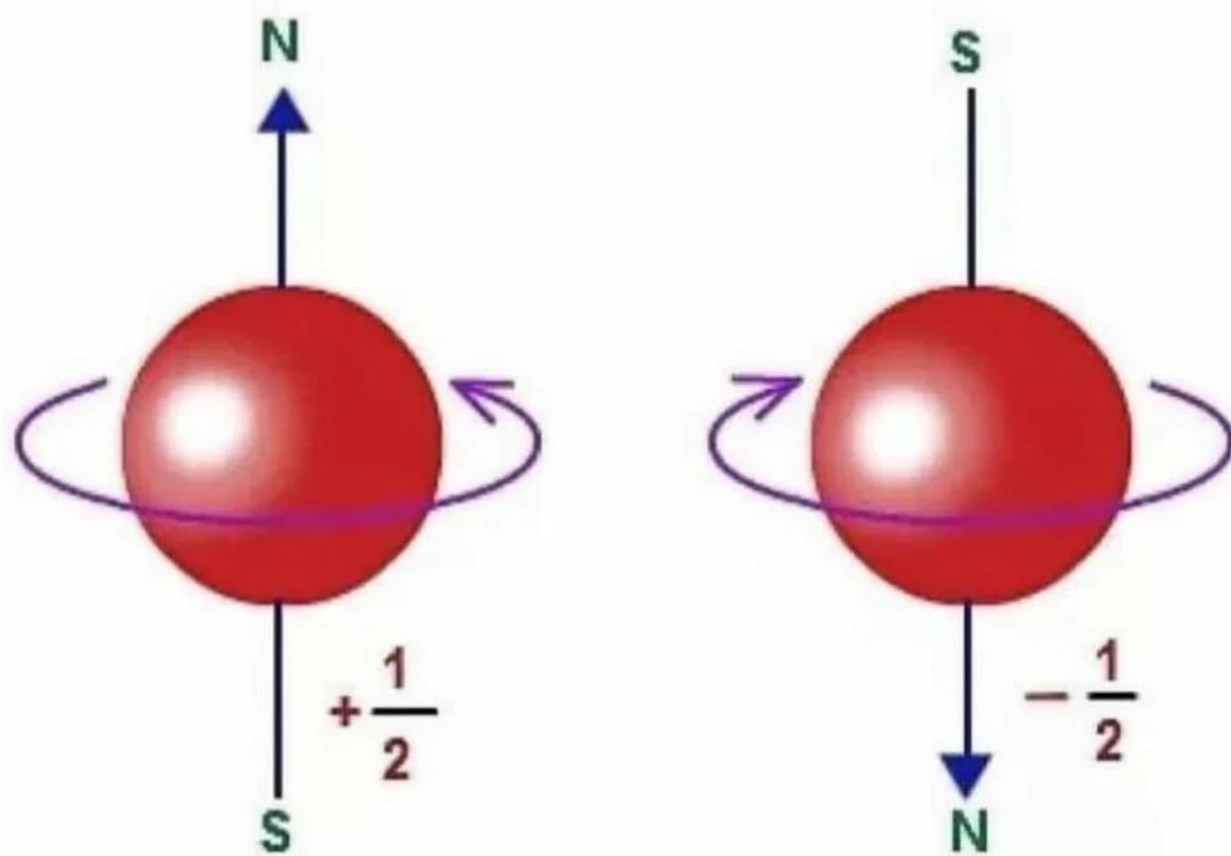
3) **Non-equilibrium situation:**

- In addition to creating more baryons than anti-baryons
- ... we need a non-equilibrium, such that we do not destroy the baryons at the same rate as they are created

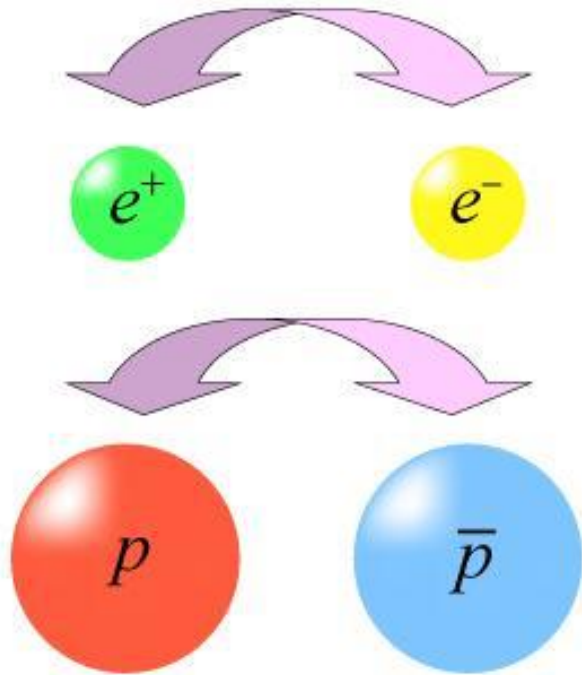


A. D. Sakharov, 1967

Electron spin explained: imagine a ball that's rotating, except it's not a ball and it's not rotating



Ladungskonjugation C

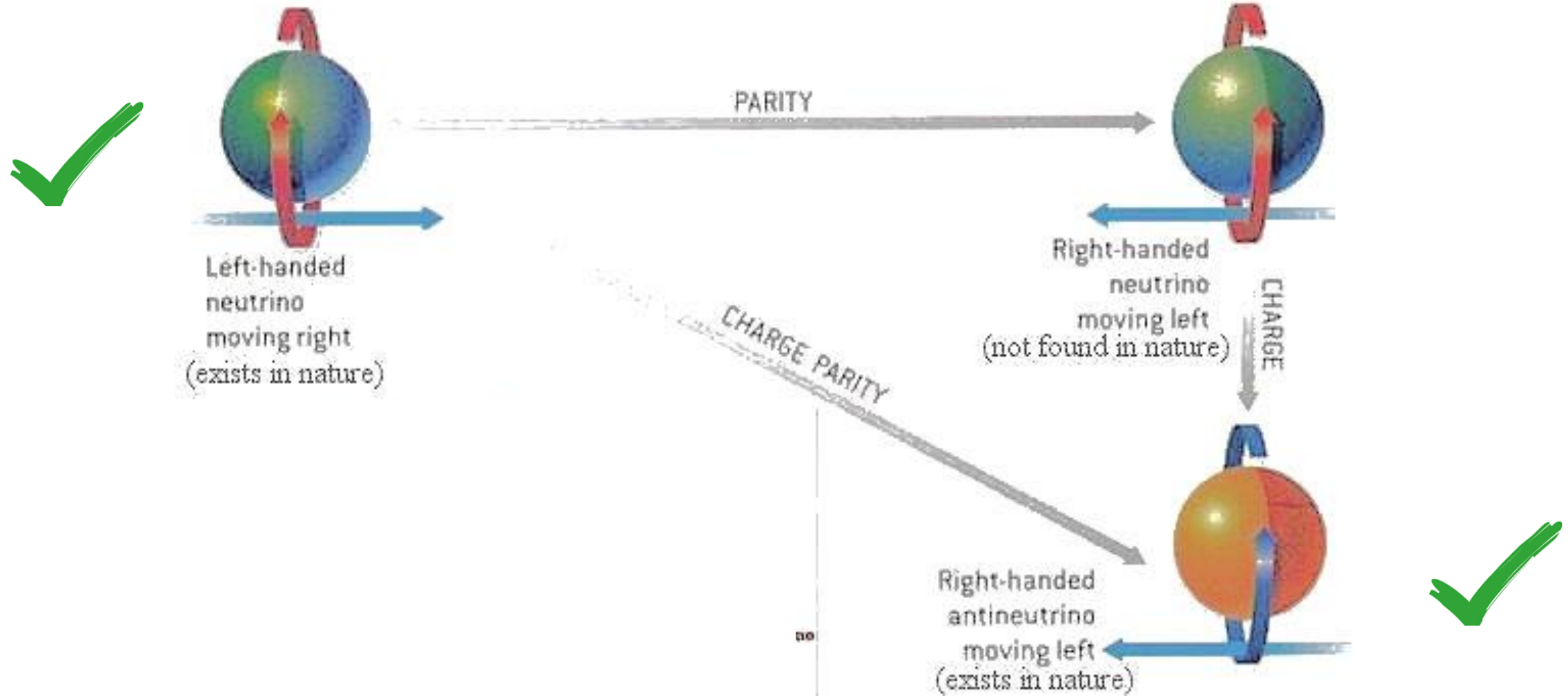


Erhalten in der elektro-magnetischen
und starken Wechselwirkung!



(Maximale) Verletzung in der
schwachen Wechselwirkung!

CP (Ladungskonjugation + Parität)



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